

Smart Metro Ticket Booking System

IDEATION PHASE

Date	
Team ID	
Project Name	Smart Metro Ticket Booking System
Maximum Marks	2 Marks

3. Brainstorming and Idea Prioritization

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3.1 Objective

The team held a brainstorming session to surface potential solution ideas for the challenge of manual, queue-based metro ticket booking. The resulting ideas were then clustered, assessed, and ranked according to how feasible they were on the ServiceNow platform, the impact on customers, and the effort needed to implement them.

3.2 Raw Brainstormed Ideas

- A digital ticket-booking portal on the ServiceNow Service Catalog, with station and travel-date pickers.
- An automated fare engine driven by Flow Designer, based on source/destination distance and passenger count.
- QR-code digital tickets for scanning at entry/exit gates.
- SMS, email, and push notifications confirming bookings and delivering tickets.
- Wallet, UPI, and card payment gateway integration for instant, cashless payment.
- Season-pass and monthly-pass booking with auto-renewal reminders.
- A live queue/crowd status view built into the booking catalog item.
- A self-service workflow for ticket cancellation and refunds.
- A chatbot / virtual agent to assist with booking and answer FAQs.
- An admin dashboard for metro staff to track bookings, revenue, and traffic by station.
- An accessibility mode with voice guidance for elderly and differently-abled commuters.
- A group-booking option for families or corporate travel.
- Integration with gate hardware/IoT scanners for automatic QR verification.
- An analytics and reporting module for predicting peak hours and optimizing fares.

3.3 Idea Grouping (Affinity Mapping)

Theme	Grouped Ideas
Core Booking Experience	Service Catalog booking form; station/date/passenger selection
Communication & Delivery	Notifications (SMS/Email/Push); booking confirmation
Payments	UPI/Card/Wallet integration; season-pass renewal
Operations & Admin	Admin dashboard; analytics/reporting; gate hardware integration
Support & Accessibility	Virtual agent/chatbot; voice-assisted accessibility mode; cancellation

3.4 Prioritization Matrix (Impact vs. Effort)

Idea	Customer Impact	Implementation Effort	Priority
Service Catalog based ticket booking form	High	Low	P0 - Must Have
Automated fare calculation (Flow Designer)	High	Medium	P0 - Must Have
QR-code digital ticket generation	High	Low	P0 - Must Have
Booking confirmation notifications (SMS/Email)	High	Low	P0 - Must Have

Idea	Customer Impact	Implementation Effort	Priority
Payment gateway integration	High	Medium	P1 - Should Have
Ticket cancellation/refund workflow	Medium	Medium	P1 - Should Have
Admin dashboard for staff/operations	Medium	Medium	P1 - Should Have
Season pass booking & renewal	Medium	Medium	P2 - Could Have
Virtual Agent/chatbot support	Medium	High	P2 - Could Have
Gate hardware/IoT scanner integration	Medium	High	P2 - Could Have
Accessibility voice-assistance mode	Medium	High	P2 - Could Have
Analytics & peak-hour prediction	Low	High	P3 - Future Scope

3.5 Finalized Ideas for Phase 1 (MVP)

Guided by the prioritization matrix, the following ideas were locked in as the Minimum Viable Product (MVP) scope for this project — all deliverable using core ServiceNow components:

- A Service Catalog item for choosing source station, destination station, travel date, and passenger count.
- A Flow Designer workflow that automatically works out the fare from station distance/zone and passenger count.
- Automatic issuance of a unique digital ticket with an embedded QR code once booking succeeds.
- A notification module that instantly sends the digital ticket and booking confirmation to the commuter.
- Basic tracking of booking history/status for each user via the Service Catalog request record.